Lecture 1
Probability and Bayesian Statistics

From Bayesian statistics to probabilistic machine learning

Probabilistic Machine Learning 2027-1

2026-10-17

Section 1

A familiar modeling language

The Beta-Binomial model

Observed

the coin tosses $y_1, ..., y_N$

Unknown

the coin bias θ

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y_i \sim \text{Bernoulli}(\theta), \quad i = 1, ..., N$$

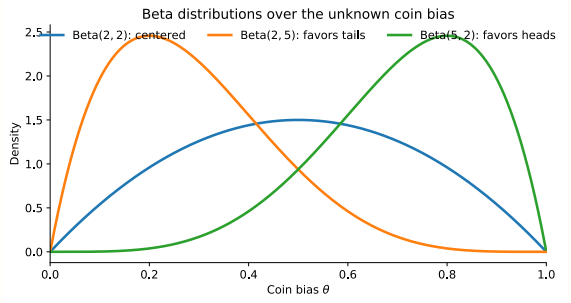
Code heads as $y_i = 1$ and tails as $y_i = 0$

$$N_1 = \sum_{i=1}^{N} y_i, \quad N_0 = N - N_1$$

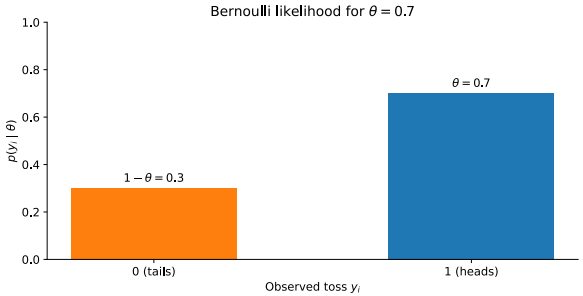
- A statistical model has data and unknown parameters
- How could observed data (y) have been generated.
- What does it tell us about the parameters (θ)
- N_1 and N_0 count heads and tails in the observed sample

Models are built out of distributions

$$\theta \sim \text{Beta}(\alpha, \beta)$$



$$y_i \sim \text{Bernoulli}(\theta)$$
$$i = 1, \ldots, N$$

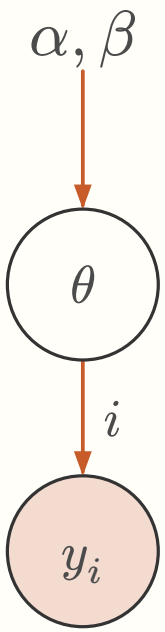


- Beta for continuous variables on $[0, 1]$.
- Given θ: independent Bernoulli tosses; Binomial for the count N_1.
- Distributions are parameterized families

Models can be drawn as graphs

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y_i \sim \text{Bernoulli}(\theta)$$



- Nodes represent quantities.
  - Shading marks observed data.
- Arrows encode conditional dependence.
  - We will not use Kruschke diagrams

In this course, graphs will allow us to factor complex problems into smaller, more tractable problems

Bayes' rule combines prior and likelihood

Carried forward Earlier in this section: Beta–Bernoulli model

Model

$$\theta \sim \text{Beta}(\alpha, \beta)$$
$$y_i \sim \text{Bernoulli}(\theta)$$

Bayes' rule

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}$$
$$\propto p(y \mid \theta)p(\theta)$$

- Bayes' rule combines prior and likelihood into a posterior
  - **Prior:** $p(\theta)$; **Likelihood:** $p(y \mid \theta)$; **Posterior:** $p(\theta \mid y)$
- We explored the posterior of θ, treating $p(y)$ as a normalizing constant

Conjugacy permits simple calculations

Carried forward Earlier in this section: model and Bayes' rule

Model

$$\theta \sim \text{Beta}(\alpha, \beta)$$
$$y_i \sim \text{Bernoulli}(\theta)$$

Bayes rule

$$p(\theta \mid y) \propto p(y \mid \theta) p(\theta)$$

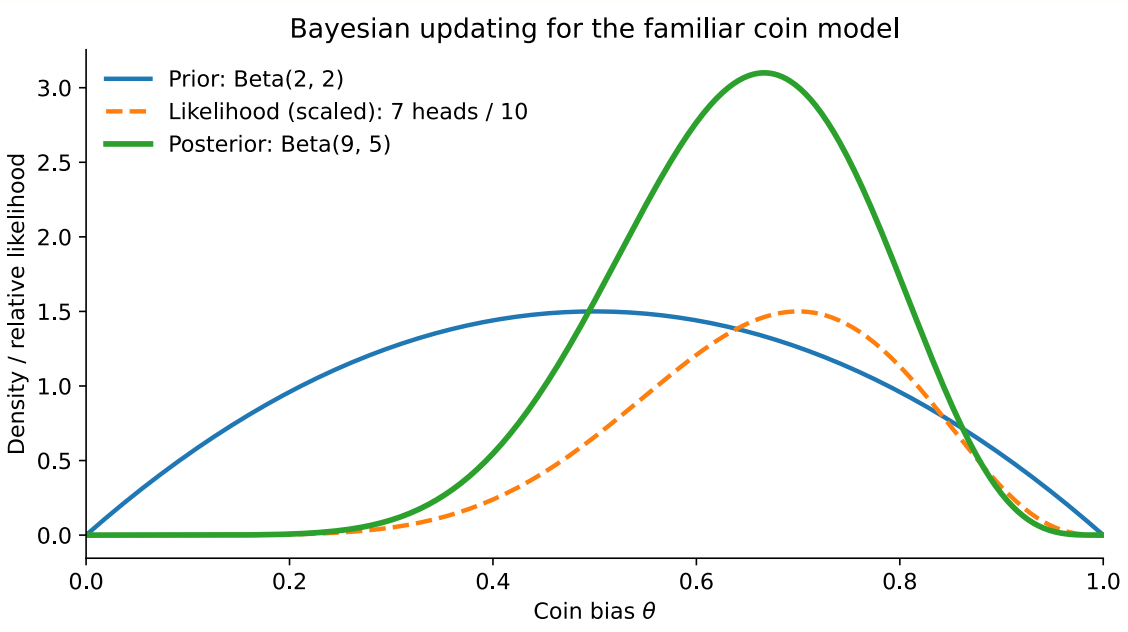
Applying Bayes' rule to our model

Prior	$p(\theta) = \text{Beta}(\alpha, \beta)$
Likelihood	$p(y \mid \theta) \propto \theta^{N_1}(1-\theta)^{N_0}$
Posterior	$p(\theta \mid y) = \text{Beta}(\alpha + N_1, \beta + N_0)$

Conjugacy

The prior and likelihood are conjugate when the posterior belongs to the same distribution family as the prior.

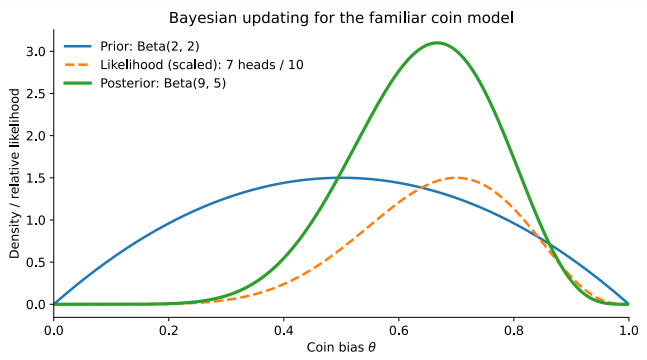
The posterior is a distribution representing uncertainty



Bayesian updating for the familiar coin model

- The prior and the posterior are both probability distributions
- Posterior width represents remaining uncertainty
- In this example, the posterior is narrower than the prior

Summary: A familiar modeling language



TAKE HOME

- A statistical model has data and unknown parameters
- Bayes' rule combines prior and likelihood into a posterior
- Posterior width represents remaining uncertainty

Section 2

From statistics to machine-learning problems

Same probabilistic language, different organizing questions

Statistics focuses on the unknown parameters

$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{p(D)}$$

- Get data → build a model → apply Bayes' rule → interpret θ
  - For many scientific questions, that posterior **is** the answer
- Statistics arose from an interest in the underlying parameters in nature

Machine learning grew out of computer science



- A computational task is an **input/output contract**
  - Produce a function f that maps input x to output y
- Implementation is secondary to the requirement
- Multiple implementations may be possible
- Machine learning developed in this context

Early machine learning grew out of classification problems

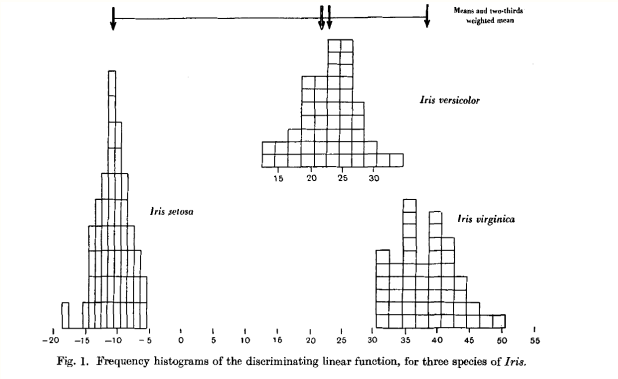


Fig. 1. Frequency histograms of the discriminating linear function, for three species of *Iris*.

Fisher (1936), Fig. 1, *The Use of Multiple Measurements in Taxonomic Problems*

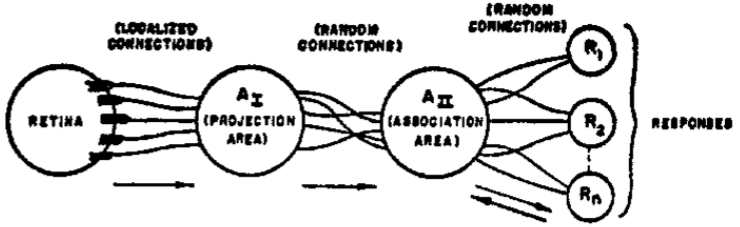
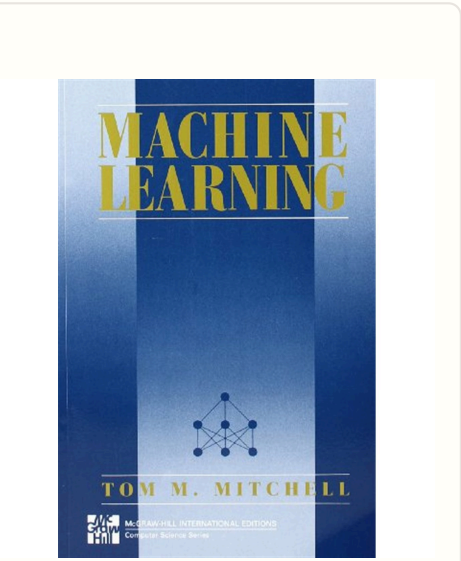


FIG. 1. Organization of a perceptron.

Rosenblatt (1958), Fig. 1, *The Perceptron*

- Fisher's discriminant and Rosenblatt's perceptron share an interface
  - complex input → class or response
- Machine learning asks how data can produce the required function f

Machine learning defined



Tom M. Mitchell, *Machine Learning* (1997), cover

A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.

Training	**Task**	**Evaluation**
$f_T : \mathrm{E} \rightarrow f_A$	$f_A : \mathrm{In} \rightarrow \mathrm{Out}$	$f_P : (\mathrm{In}, \mathrm{Out}) \rightarrow \mathrm{P}$
E: Experience	f_A: Task function	P: Performance

- Machine learning is a mapping from $(E, f_P) \rightarrow f_T \rightarrow f_A$
  - That maximizes f_A performance according to P

Supervised experience supports prediction

Experience: paired examples

$$f_T : \mathrm{E} \quad \to \quad f_A$$

$$f_T : D = \{(x_i, y_i)\}_{i=1}^{N} \quad \to \quad f_D$$

paired examples → learned predictor

Task: prediction

$$f_A : \mathrm{In} \quad \to \quad \mathrm{Out}$$

$$f_D : x_* \quad \to \quad \hat{y}_*$$

new input → predicted output

- Experience is a training set $D = \{(x_n, y_n)\}$
  - x: called feature, covariate, or predictor
  - y: called label, target, or response
- Task is to produce a predictor f_D that maps new x_* to correct y_*
- Performance is measured by a loss, introduced shortly

Classification: Iris flowers



Setosa



Versicolor



Virginica

Murphy (2025), Book 1, Fig. 1.1

Table 1.1: subset of the Iris design matrix

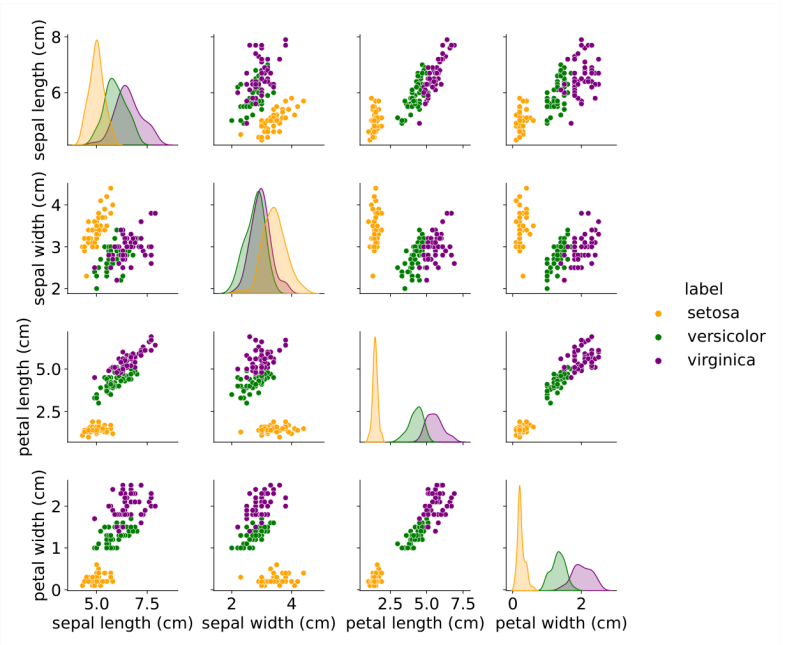
index	sl	sw	pl	pw	label
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
dots	dots	dots	dots	dots	dots
50	7.0	3.2	4.7	1.4	Versicolor
dots	dots	dots	dots	dots	dots
149	5.9	3.0	5.1	1.8	Virginica

sl: sepal length; sw: sepal width; pl: petal length; pw: petal width

Murphy (2025), Book 1, Table 1.1

- Classification: $y \in \{1, \ldots, C\}$
  - The classes are unordered and mutually exclusive
- For the Iris dataset
  - $y \in \{\text{Setosa}, \text{Versicolor}, \text{Virginica}\}$
  - $x \in \mathbb{R}^4$

Exploratory data analysis



Murphy (2025), Book 1, Fig. 1.3; generated by `iris_plot.ipynb`

- Training set
  - 150 labeled examples
  - 50 in each class
- Pair plots shows every pair of features
  - Diagonal shows marginal distributions
- Always inspect data before modeling
  - Structure
  - Outliers
  - Missing values

Performance: empirical risk minimization

$$L(x, y; \theta) = \ell(y, f(x; \theta))$$

$$L(\theta) = \frac{1}{N} \sum_{n=1}^{N} L(x_n, y_n; \theta)$$

$$\hat{\theta} = \arg\min_{\theta} L(\theta)$$

Given samples $D = \{x^1, ..., x^N\}$ from $p(X)$, where $X \in \mathbb{R}$, the empirical distribution is $\hat{p}_N(x) = \frac{1}{N} \sum_{n=1}^{N} \delta_{x^n}(x)$.

- The loss function: the cost of predicting $f(x; \theta)$ when the true label is y
- The empirical risk: loss evaluated on the training set
- Empirical risk minimization: minimizing the empirical loss

A loss function

$$L(x, y; \theta) = \ell(y, f(x; \theta))$$

$$\ell(y, f(x; \theta)) = \ell_{01}(y, \hat{y}) = \mathbb{I}(y \neq \hat{y})$$

$$\mathbb{I}(e) = \begin{cases} 1 \text{ if } e \text{ is true} \\ 0 \text{ if } e \text{ is false} \end{cases}$$

- A loss function assigns a cost to a prediction when the true label is known
- The zero-one loss is one for a misclassification and zero for a correct classification
- $\mathbb{I}(e)$ is an indicator function: it evaluates whether condition e holds

A probabilistic perspective

$$p(x \mid y = c, \theta)$$

$$p(y = c \mid x, \theta) \propto p(x \mid y = c, \theta) p(y = c)$$

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} p(y = c \mid x; \theta)$$

- The exploratory data analysis suggests:
  - Features in each class have similar distributions
  - Different classes have different distributions
- Bayes' rule converts those ingredients to posterior class probabilities
- A possible classifier: choose the class with maximum a posteriori probability

Task, experience, and performance

Task

What must the learned system do?

$$x \in \mathbb{R}^4 \to \hat{y} \in \mathcal{Y}$$

four measurements → species

Experience

What data, feedback, or interaction is available?

$$D = \{(x_n, y_n)\}_{n=1}^{N}$$

150 labeled flowers

Performance

How is success evaluated?

$$L(\theta) = \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}(y_n \neq f(x_n; \theta))$$

zero-one loss / misclassification rate

- A learning problem is defined by three ingredients: task, experience, and performance
  - **Task:** predict the Iris species from four sepal and petal measurements
- **Experience:** 150 labeled flowers, with 50 examples from each species
- **Performance:** mean zero-one loss, the training misclassification rate

Summary: From statistics to machine learning

Training

$f_T : \mathrm{E} \to f_A$
E: experience

Task

$f_A : \mathrm{In} \to \mathrm{Out}$
f_A: learned function

Evaluation

$f_P : (\mathrm{In}, \mathrm{Out}) \to P$
P: performance

TAKE HOME

- Machine learning asks how data can produce the required function f
- A learning problem is defined by three ingredients: task, experience, and performance

Section 3

Classes of machine learning problems

Experience regimes, task classes, and performance criteria

Experience taxonomy

Supervised

$$D = \{(x_n, y_n)\}$$

Unsupervised

$$D = \{x_n\}$$

Self-supervised

$$D = \{x_n, f(x_n)\} \text{ or } \{f(x_n), x_n\}$$

Reinforcement / interaction

$$\begin{aligned} \text{decision} &: x(t) \to a(t) \\ \text{action} &: (x(t), a(t)) \to x(t+1) \\ \text{reward} &: x(t+1) \to r(t+1) \end{aligned}$$

> Experience describes what the learner gets to observe

Task taxonomy

Prediction

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

Generation

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

Discovery

Task: $\{x_n\} \to (\{z\}, f_D(x)) \to \hat{z} = f_D(x_*)$

Performance: $f_P(\hat{z}, x_*)$

Action

Task: $D \to a_t \sim \pi_D(\cdot \mid x_t)$

$(x_t, a_t) \overset{\text{environment}}{\to} (x_{t+1}, r_{t+1})$

Performance: $f_P(\{r_t\})$

Task describes what the learned system must do.

Prediction	Map inputs to outputs
Generation	Produce new examples
Discovery	Find a representation of the data
Action	Choose actions to achieve desired outcomes

There are many combinations of experience and task

Experience	Prediction	Generation	Discovery	Action
Supervised	Classification	—	—	Imitation learning
Unsupervised	—	Density modeling	Clustering / PCA	—
Self-supervised	Next-token prediction	Language generation	Representation learning	—
Reinforcement / interaction	—	—	—	Policy learning

- Not every type of experience supports every task.
- Common pairings do exist.

Summary: Two taxonomies, three ingredients

Experience: what the learner observes

Supervised
$D = \{(x_n, y_n)\}$

Unsupervised
$D = \{x_n\}$

Self-supervised
$x_n, f(x_n)$

Interaction
$x_t \to a_t \to r_{t+1}$

Task: what the system must do

Prediction
$x_* \to \hat{y}_*$

Generation
$x_* \sim p_D(x|c)$

Discovery
$x_* \to \hat{z}_*$

Action
$x_t \to a_t$

TAKE HOME

- Experience describes what the learner gets to observe
- Task describes what the learned system must do
- Not every type of experience supports every task; common pairings do exist

Section 4

From datasets to learning problems

The same dataset can pose different learning problems

Popular datasets

Iris





AGE
TASKS

Since 1936 – about 90 years
species prediction; clustering; visualization

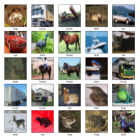
MNIST



AGE
TASKS

Since 1998 – about 28 years
digit classification; representation learning; generation

CIFAR-10



AGE
TASKS

Since 2009 – about 17 years
object classification; robustness; generation

IMDB movie reviews

1. this film was just brilliant casting location scenery story direction everyone's really suited the part they played robert <UNK> is an amazing actor ...
2. big hair big boobs bad music and a giant safety pin these are the words to best describe this terrible movie i love cheesy horror movies and i've seen hundreds...

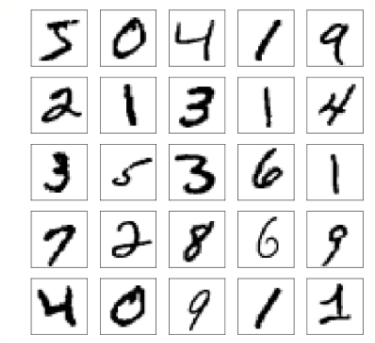
AGE
TASKS

Since 2011 – about 15 years
sentiment classification; text representation

Murphy (2025), Book 1, Figs. 1.1, 1.12a, 1.13b, and Table 1.3

- Machine learning uses common datasets for practice and example problems
- The same dataset can support different learning tasks
- Datasets support benchmarks that make comparisons easy

MNIST: the same taxonomy, one dataset



$$x_n \in \{0, \ldots, 255\}^{28 \times 28}, \quad y_n \in \{0, \ldots, 9\}$$

Murphy (2025), Book 1, Fig. 1.12a

Prediction

$$\hat{y} = f_D(x_*)$$

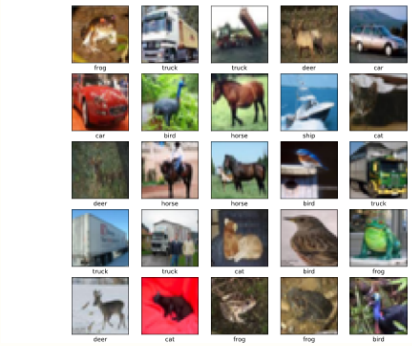
Discovery

$$\hat{z} = f_D(x_*)$$

Generation

$$x_* \sim p_D(x \mid c)$$

CIFAR-10: higher-dimensional inputs, the same questions



$$x_n \in \mathbb{R}^{32\times32\times3}, \quad y_n \in \{1, \ldots, 10\}$$

Murphy (2025), Book 1, Fig. 1.13b

Prediction

$$\hat{y} = f_D(x_*)$$

$$L(\theta; D_{\text{train}}) \xrightarrow{D_{\text{train}} \to D_{\text{test}}} L(\theta; D_{\text{test}})$$

IMDB reviews: text as sequence data

1. this film was just brilliant casting location scenery story direction everyone's really suited the part they played robert <UNK> is an amazing actor ...
2. big hair big boobs bad music and a giant safety pin these are the words to best describe this terrible movie i love cheesy horror movies and i've seen hundreds...

Murphy (2025), Book 1, Table 1.3

Prediction

$$\hat{y} = f_D(x_*)$$

Discovery

$$\hat{z} = f_D(x_*)$$

Generation

$$w_t \sim p_D(w_t \mid w_{<t})$$

Learning problems become probabilistic queries

Prediction

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

Generation

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

Discovery

Task: $\{x_n\} \to (\{z\}, f_D(x)) \to \hat{z} = f_D(x_*)$

Performance: $f_P(\hat{z}, x_*)$

Action

Task: $D \to a_t \sim \pi_D(\cdot \mid x_t)$

$(x_t, a_t) \overset{\text{environment}}{\to} (x_{t+1}, r_{t+1})$

Performance: $f_P(\{r_t\})$

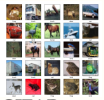
- A learning problem is specified by the variables available and the query we want answered
- Performances are expectations over future cases: $f_P = E[\cdot]$
- Probabilistic notation gives a common language for expressing those queries

Summary: From datasets to learning problems

Common practice datasets



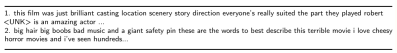
Iris



CIFAR-10



MNIST



IMDB

One dataset, different tasks



Prediction

$\hat{y} = f_D(x_*)$

Discovery

$\hat{z} = f_D(x_*)$

Generation

$x_* \sim p_D(x|c)$

TAKE HOME

- Machine learning uses common datasets for practice and example problems
- The same dataset can support different learning tasks
- A learning problem is specified by the variables available and the query we want answered
- Probabilistic notation gives a common language for expressing those queries

Summary: Probability for machine learning

Joint distribution: $p_D(x_{\text{obs}}, z, y_*)$

Summarize
$E[g(X)] = \int g(x)p(x)\,\mathrm{d}x$

Marginalize
$p(x) = \int p(x, y)\,\mathrm{d}y$

Condition
$p(x|y) = \frac{p(x,y)}{p(y)}$

TAKE HOME

- The task determines which quantity we need from the joint distribution
- A small set of recurring operations—summarize, marginalize, condition—answers many of those questions
- Integrals are a central tool for answering these queries

Section 5

Joint Gaussian distributions and geometry

Describe and transform joint uncertainty with a mean and covariance

Gaussians: a concrete example

One variable

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}}$$

***D* coordinates**

$$\mathcal{N}(\boldsymbol{x} \mid \mu, \Sigma) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\mu)^T\Sigma^{-1}(\boldsymbol{x}-\mu)\right)}{(2\pi)^{\frac{D}{2}}\ |\Sigma|^{\frac{1}{2}}}$$

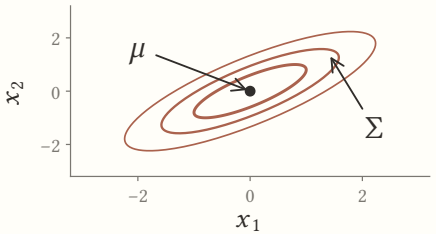
Murphy, Book 2 §2.3.1.1, Eq. 2.71; scalar case: $D = 1$

- One variable: mean μ and positive variance $\sigma^2 > 0$
- Mean $\mu \in \mathbb{R}^D$; these densities and inverse formulas assume covariance $\Sigma \in \mathbb{R}^{D\times D}$ with $\Sigma \succ 0$
- Gaussian means, marginals, and conditionals have closed forms

The three operations in multivariate Gaussian systems

One joint Gaussian: $p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x} \mid \mu, \Sigma)$

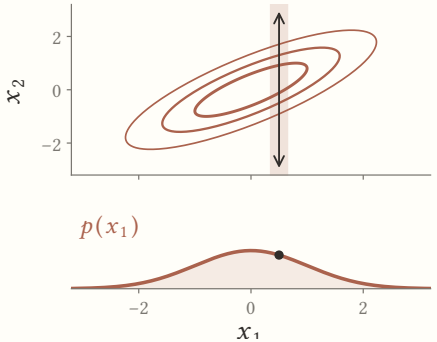
Summarize



Center and spread

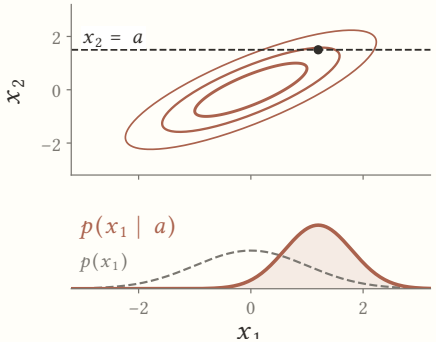
$$\mu = \int \boldsymbol{x} p(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x}$$

Marginalize



$$p(x_1) = \int p(x_1, x_2) \, \mathrm{d}x_2$$

Condition



$$p(x_1 \mid x_2 = a) = \frac{p(x_1, a)}{\int p(u, a) \, \mathrm{d}u}$$

Course figures generated by `gaussian_geometry_figures.ipynb`

- **Summarize:** describe the center and spread with μ and Σ
- **Marginalize:** integrate over the coordinate we ignore
- **Condition:** fix an observed coordinate and normalize the slice

Covariance summarizes joint variation

For a vector $\boldsymbol{x}$ with mean $\mu = E[\boldsymbol{x}]$

$$\Sigma_{ij} = \operatorname{Cov}[X_i, X_j]$$
$$= E[(X_i - \mu_i)(X_j - \mu_j)]$$

$$\Sigma = E[(\boldsymbol{x} - \mu)(\boldsymbol{x} - \mu)^T]$$
$$= E[\boldsymbol{x}\boldsymbol{x}^T] - \mu\mu^T$$

For three coordinates

$$\Sigma = \begin{bmatrix} V[X_1] & \operatorname{Cov}[X_1, X_2] & \operatorname{Cov}[X_1, X_3] \\ \operatorname{Cov}[X_2, X_1] & V[X_2] & \operatorname{Cov}[X_2, X_3] \\ \operatorname{Cov}[X_3, X_1] & \operatorname{Cov}[X_3, X_2] & V[X_3] \end{bmatrix}$$

Murphy, Book 1 §3.1.1, Eqs. 3.1–3.4

- Positive / negative covariance: deviations tend to have the same / opposite signs
- Diagonal entries are variances; off-diagonal entries are covariances
- Zero covariance does not imply independence

Correlation separates direction from units

$$\rho_{ij} = \frac{\mathrm{Cov}[X_i, X_j]}{\sigma_i \sigma_j}, \quad -1 \le \rho_{ij} \le 1$$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 & \rho_{13}\sigma_1\sigma_3 \\ \rho_{12}\sigma_1\sigma_2 & \sigma_2^2 & \rho_{23}\sigma_2\sigma_3 \\ \rho_{13}\sigma_1\sigma_3 & \rho_{23}\sigma_2\sigma_3 & \sigma_3^2 \end{bmatrix}$$

Murphy, Book 1 §3.1.2, Eq. 3.7; Book 2 §2.3.1.1, Eq. 2.73

- ρ_{ij} describes the direction and strength of linear association
- $\sigma_1, \sigma_2, \sigma_3$ set the coordinate scales

Affine maps carry Gaussian means and covariances

$$\boldsymbol{y} = A\boldsymbol{x} + b$$

$$E[\boldsymbol{y}] = A\mu + b$$

$$\mathrm{Cov}[\boldsymbol{y}] = A\Sigma A^T$$

Covariance proof

Center: $\boldsymbol{d} = \boldsymbol{x} - \mu$

$$\begin{aligned}
\boldsymbol{y} &= A(\boldsymbol{d} + \mu) + b \\
&= A\boldsymbol{d} + (A\mu + b) \\
\boldsymbol{y} - E[\boldsymbol{y}] &= A\boldsymbol{d} \\
\mathrm{Cov}[\boldsymbol{y}] &= E\big[(A\boldsymbol{d})(A\boldsymbol{d})^T\big] \\
&= AE\big[\boldsymbol{d}\boldsymbol{d}^T\big]A^T \\
&= A\Sigma A^T
\end{aligned}$$

Murphy, Book 1 §3.1.1, Eq. 3.5

- Gaussianity is preserved; a shift changes only the mean
- A linear map changes the scale, orientation, and dimension of variation

Mahalanobis distance uses the cloud's own scale

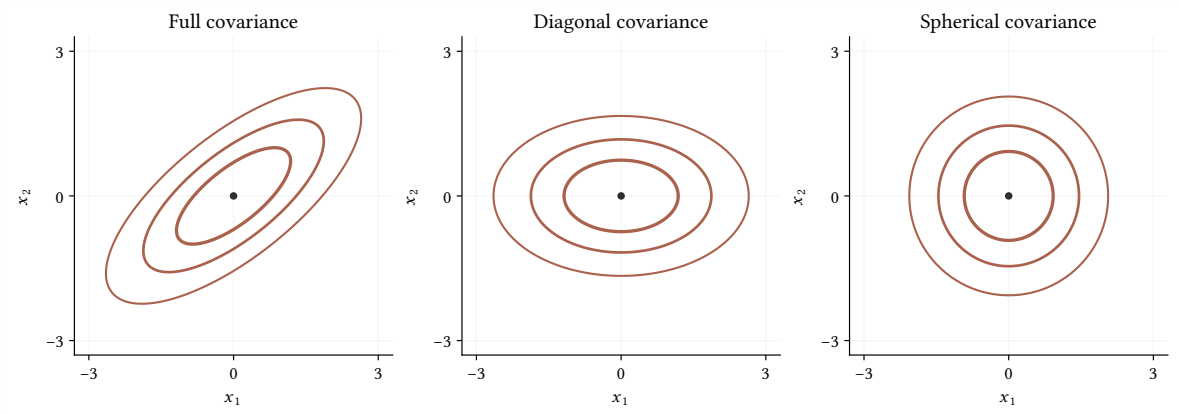
$$d_\Sigma(\boldsymbol{x}, \mu)^2 = (\boldsymbol{x} - \mu)^T \Sigma^{-1} (\boldsymbol{x} - \mu)$$

$$\Sigma = \sigma^2 I_D \quad \rightarrow \quad d_\Sigma^2 = \frac{\|\boldsymbol{x} - \mu\|^2}{\sigma^2}$$

Murphy, Book 2 §2.3.1.1, Eq. 2.72

- Equal Mahalanobis distances trace equal-density contours
- With spherical covariance, it becomes squared Euclidean distance in standard-deviation units

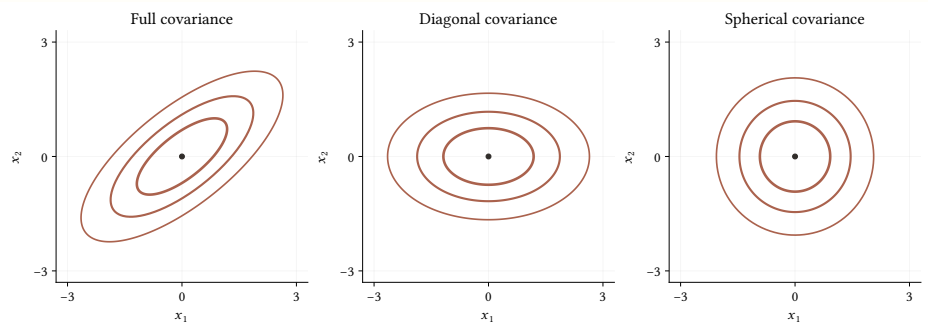
Covariance controls the contour shape



Course rendering of Murphy, Book 2, Fig. 2.8

- A full covariance can rotate the principal axes
- Diagonal covariance removes linear correlation in these coordinates
- Spherical covariance assigns the same variance in every direction

Summary: Joint Gaussian geometry



Affine map

$$\boldsymbol{y} = A\boldsymbol{x} + b$$
$$E[\boldsymbol{y}] = A\mu + b$$
$$\mathrm{Cov}[\boldsymbol{y}] = A\Sigma A^T$$

TAKE HOME

- Mean vector μ locates the cloud; covariance Σ sets its shape
- A linear map changes the scale, orientation, and dimension of variation

Section 6

Marginals and conditionals of a Gaussian

Ignore coordinates or use observations to update uncertainty

Partition the variables and their parameters together

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}, \quad \Sigma_{21} = \Sigma_{12}^T$$

Murphy, Book 2 §2.3.1.3, Eq. 2.75

- x_1 and x_2 can each be vectors
- The diagonal blocks describe within-group variation
- The cross blocks describe variation between groups

Marginalization keeps the matching blocks

$$x \sim \mathcal{N}(\mu, \Sigma)$$

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$

$$= \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11})$$

$$p(x_2) = \mathcal{N}(x_2 \mid \mu_2, \Sigma_{22})$$

Marginal proof

Select x_1: $P = [I \;\; 0]$, $\quad x_1 = Px$

By the affine rule:

$$x_1 \sim \mathcal{N}(P\mu, P\Sigma P^T)$$

$$P\mu = \mu_1$$

$$P\Sigma P^T = [I \;\; 0] \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \begin{bmatrix} I \\ 0 \end{bmatrix}$$

$$= \Sigma_{11}$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.76–2.77; affine rule: Book 1 §3.1.1, Eq. 3.5

- Marginalizing x_2 integrates over all its possible values
- Cross-covariance affects conditioning, not the marginal of x_1

Conditioning shifts the mean by an observed deviation

$$p(x_1 \mid x_2) = \mathcal{N}\left(x_1 \mid \mu_{1|2}, \Sigma_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

Mean proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2} = x_1 - \mu_1 - B(x_2 - \mu_2)$$

$$E[r] = 0,$$

$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22}$$

$$= 0$$

(r, x_2) is jointly Gaussian:

zero cross-covariance gives independence

$$E[r \mid x_2] = E[r] = 0$$

$$E[x_1 \mid x_2] = \mu_{1|2}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; proof uses affine Gaussian closure

- Conditioning shifts the mean according to the observed deviation $x_2 - \mu_2$, cross-covariance, and observed variance

Conditioning removes explained covariance

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

Covariance proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2}$$

From the mean proof: r and x_2 are independent

Given x_2, the mean $\mu_{1|2}$ is constant

$$\Sigma_{1|2} = \mathrm{Cov}[x_1 \mid x_2] = \mathrm{Cov}[r]$$

$$\Sigma_{11} = B\Sigma_{22}B^T + \mathrm{Cov}[r]$$

$$\Sigma_{1|2} = \Sigma_{11} - B\Sigma_{22}B^T$$

$$= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.78–2.79; proof reuses the independent residual

- Conditioning cannot increase variance in any direction
- In a Gaussian, this covariance depends on which variables are observed, not their values

The bivariate formula gives intuition

$$x_1, x_2 \in \mathbb{R}$$

$$(X_1 \mid X_2 = x_2) \sim \mathcal{N}\left(\mu_{1|2}, \sigma^2_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$$

$$\sigma_{1|2} = \sigma_1 \sqrt{1 - \rho^2}$$

Murphy, Book 1 §3.2.4; Book 2 §2.3.1.3

- Positive ρ moves the estimate in the observed direction
- $\rho = 0$ leaves both mean and variance unchanged
- Larger $|\rho|$ leaves less conditional variance

One observed coordinate can change both the estimate and its uncertainty

Supplied joint model

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}\left(\mu = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}\right)$$

Recorded: $X_2 = 1.5$; **missing:** X_1

$X_2 = 1.5$ **recorded**

$$(X_1 \mid X_2 = 1.5) \sim \mathcal{N}(0.8(1.5), 1 - 0.8^2) = \mathcal{N}(1.2, 0.36)$$

X_2 **unobserved**

$$X_1 \sim \mathcal{N}(0, 1)$$

Course-constructed example; Gaussian conditioning: Murphy, Book 2 §2.3.1.3, Eq. 2.78

- With X_2 unobserved, marginalize it to obtain $p(X_1)$.
- Both calculations use supplied μ, Σ; neither fits them.
- Compared with the marginal, conditioning raises the mean to 1.2 and reduces the variance to 0.36.

The Schur complement is the residual covariance

Conditional Gaussian

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$S = \Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$
$$(x_1 \mid x_2) \sim \mathcal{N}\big(\mu_{1|2}, S\big)$$

Centered residual

$$r = x_1 - \mu_{1|2}$$
$$r \sim \mathcal{N}(0, S)$$

The residual r is independent of x_2

Remove the part predicted by x_2:

$$\mathrm{Cov}\left[\begin{pmatrix} r \\ x_2 \end{pmatrix}\right] = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

Murphy, Book 2 §2.3.1.5, Eqs. 2.91, 2.102–2.103

- S measures the uncertainty left after subtracting the conditional mean
- Block elimination leaves S: the **Schur complement of Σ_{22} in Σ**

The joint quadratic separates into conditional and marginal parts

$$p(x_1, x_2) = p(x_1 \mid x_2)p(x_2)$$

$$(\boldsymbol{x} - \mu)^T \Sigma^{-1} (\boldsymbol{x} - \mu)$$

$$= r^T S^{-1} r + (x_2 - \mu_2)^T \Sigma_{22}^{-1} (x_2 - \mu_2)$$

Murphy, Book 2 §2.3.1.5, Eqs. 2.94–2.103

- The first quadratic describes x_1 after seeing x_2
- The second describes the marginal distribution of x_2

Summary: Marginals and conditionals of a Gaussian

Marginalize

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$
$$x_1 \sim \mathcal{N}(\mu_1, \Sigma_{11})$$

Condition

$$p(x_1|x_2) = \mathcal{N}\left(x_1|\mu_{1|2}, \Sigma_{1|2}\right)$$
$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

TAKE HOME

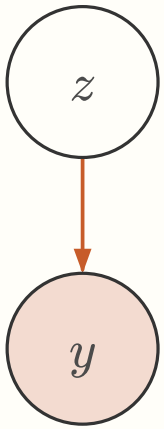
- Marginalizing x_2 integrates over all its possible values
- Conditioning shifts the mean according to the observed deviation, cross-covariance, and observed variance
- Conditioning cannot increase variance in any direction

Section 7

Linear Gaussian systems

Inference from noisy linear measurements

A linear Gaussian system is a model of state and measurement



Latent state

Observed

A prior and a measurement law

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$

$$y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$

$$p(z, y) = p(z)p(y \mid z)$$

$$y = Wz + b + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Omega)$$

Noise ε is independent of the state z.

Murphy, Book 2 §2.3.2, Eqs. 2.118–2.119

- This model specifies how an unknown state generates a noisy measurement
- The prior and noise covariances satisfy $\Sigma_z \succ 0, \Omega \succ 0$
- Here the parameters $\mu_z, \Sigma_z, W, b, \Omega$ are given

One linear Gaussian system supports different questions

$$z \sim \mathcal{N}(\mu_z, \Sigma_z), \quad y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$

Condition

$$p(z \mid y) = \frac{p(y \mid z)p(z)}{p(y)}$$

Summarize

$$E[g(z) \mid y] = \int g(z)p(z \mid y)\,\mathrm{d}z$$

Marginalize

$$p(y) = \int p(y \mid z)p(z)\,\mathrm{d}z$$

Murphy, Book 2 §2.3.2

- Conditioning asks what we can learn from an observed measurement
  - What is the distribution of the hidden state after seeing a measurement?
- Summarizing turns a distribution into a quantity
  - What is our estimate of the hidden state?
- Marginalization averages over an unknown variable to predict another
  - What is the distribution of a measurement before it is observed?

A linear Gaussian model defines a joint Gaussian

Linear Gaussian model

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$y = Wz + b + \varepsilon$$
$$\varepsilon \sim \mathcal{N}(0, \Omega)$$

ε is independent of z

Affine Gaussian rules

$$u = Ax + c$$
$$E[u] = AE[x] + c$$
$$\mathrm{Cov}[u] = A\ \mathrm{Cov}[x]A^T$$

Joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix}\right)$$
$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$
$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Murphy, Book 2 §2.3.2.1, Eq. 2.120

- The model plus affine-Gaussian rules determine the full joint distribution
- Once $p(z, y)$ is known, the Gaussian marginal and conditioning rules apply directly

Bayes' rule for Gaussians

Model and joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix} \right)$$

$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$
$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Gaussian conditioning rule

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Apply: $x_1 = z, x_2 = y$

$$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$$

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}\left(y - \mu_y\right)$$

$$\Sigma_{z|y} = \Sigma_{zz} - \Sigma_{zy}\Sigma_{yy}^{-1}\Sigma_{yz}$$

Murphy, Book 2 §2.3.2.2, Eq. 2.121

- Apply the Gaussian conditioning rule to the joint blocks
- The mean responds to observed minus predicted measurement
- The covariance subtracts the uncertainty explained by the measurement

Bayes' rule as an update formula

Carried forward Earlier: Gaussian conditioning

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}(y - \mu_y)$$
$$\mu_y = W\mu_z + b$$
$$\Sigma_{zy} = \Sigma_z W^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Define the gain

$$K = \Sigma_{zy}\Sigma_{yy}^{-1}$$
$$\mu_{z|y} = \mu_z + K(y - \mu_y)$$

Murphy, Book 2 §2.3.2.2, Eqs. 2.122–2.124

- K is the gain: it converts a measurement discrepancy into a state correction
- Posterior estimate = prior estimate + gain × prediction error

The scalar Gaussian update

Carried forward Earlier: Gaussian update formula

$$\mu_{z|y} = \mu_z + K(y - \mu_y)$$
$$K = \Sigma_{zy}\Sigma_{yy}^{-1}$$

Scalar specialization

$$y, z \in \mathbb{R}, \quad W = 1, \quad b = 0$$
$$\Sigma_z = \sigma_z^2, \quad \Omega = \sigma_\varepsilon^2$$
$$z \sim \mathcal{N}(\mu_z, \sigma_z^2)$$
$$y = z + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

ε is independent of z

Predicted mean

$$\mu_y = E[z + \varepsilon] = \mu_z$$

Cross-covariance

$$\Sigma_{zy} = \mathrm{Cov}[z, z + \varepsilon] = \sigma_z^2$$

Measurement variance

$$\Sigma_{yy} = \mathrm{Var}[z + \varepsilon] = \sigma_z^2 + \sigma_\varepsilon^2$$

$$K = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\varepsilon^2} \quad \rightarrow \quad \mu_{z|y} = \mu_z + K(y - \mu_z)$$

Murphy, Book 2 §2.3.2.2; scalar specialization of Eqs. 2.121–2.123

- The scalar gain is prior state variance divided by total measurement variance

The update is weighted by relative uncertainty

$$\mu_{z|y} = \mu_z + K(y - \mu_z), \quad K = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\varepsilon^2}$$

◆ = posterior mean $\mu_{z|y}$

Precise measurement

$\sigma_\varepsilon^2 \ll \sigma_z^2 \quad K \approx 1$



Trust the measurement

Equal uncertainty

$\sigma_\varepsilon^2 = \sigma_z^2 \quad K = \frac{1}{2}$



Split the difference

Noisy measurement

$\sigma_\varepsilon^2 \gg \sigma_z^2 \quad K \approx 0$



Trust the prior

Scalar interpretation of Murphy, Book 2 §2.3.2.2

- K is the fraction of the prediction error used in the correction
- More measurement noise gives smaller K; more prior uncertainty gives larger K

Inference often combines conditioning and summarizing

Observed		**Condition**		**Summarize**
y	$\rightarrow$	$p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z\mid y}, \Sigma_{z\mid y}\right)$	$\rightarrow$	$\hat{z} = E[z \mid y] = \mu_{z\mid y}$

Murphy, Book 2 §2.3.2.2–2.3.2.3

$$E[z \mid y] = \int z p(z \mid y)\, \mathrm{d}z$$

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z\mid y}$

Choose a reconstruction by its conditional squared-error risk

Squared-error risk

Observed y; $z, a \in \mathbb{R}^d$

$$E[\|z\|^2|y] < \infty$$

$$L(a, z) = \|z - a\|^2$$

$$R(a|y) = E[L(a, z)|y]$$

Centered error

$$m = E[z|y], \quad r = z - m$$

$$z - a = r + (m - a)$$

Squared-error expansion

$$\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2$$

- Choose a to minimize conditional expected squared error $R(a|y)$.
- Given y, offset $m - a$ is fixed; uncertainty remains in r.

Centering separates residual uncertainty from estimation error

Conditional expansion Previous slide: pointwise squared-error expansion

$$\|z-a\|^2 = \|r\|^2 + 2r^T(m-a) + \|m-a\|^2$$

Take $E[\cdot\,|y]$; $m-a$ is fixed

Cross-term

$$E[r|y] = E[z|y] - m = 0$$
$$E\left[r^T(m-a)|y\right] = E[r|y]^T(m-a) = 0$$

Residual energy

$$V = \mathrm{Cov}[z|y]$$
$$E\left[\|r\|^2|y\right] = \sum_{j=1}^{d} E\left[r_j^2|y\right] = \mathrm{tr}\, V$$

$$R(a|y) = \mathrm{tr}\, V + \|m-a\|^2$$

Murphy, Book 1 §5.1.5.1; vector decomposition: course elaboration

- Risk is residual uncertainty plus squared distance from the posterior mean.

The posterior mean minimizes squared-error risk

$$R(a|y) = \underbrace{\operatorname{tr} V}_{\text{independent of a}} + \underbrace{\|m-a\|^2}_{\text{nonnegative; zero iff } a=m}$$

Consequences at the minimizing action

Minimum risk

$$\min_a R(a|y) = \operatorname{tr} V$$

Optimal action

$$a^*(y) = m = E[z|y]$$

For the conditional Gaussian

$$\min_a R(a|y) = \operatorname{tr} \Sigma_{z|y} \qquad a^*(y) = \mu_{z|y}$$

- The conditional mean minimizes model expected squared error.
- The posterior covariance describes uncertainty that remains after the mean action is chosen.

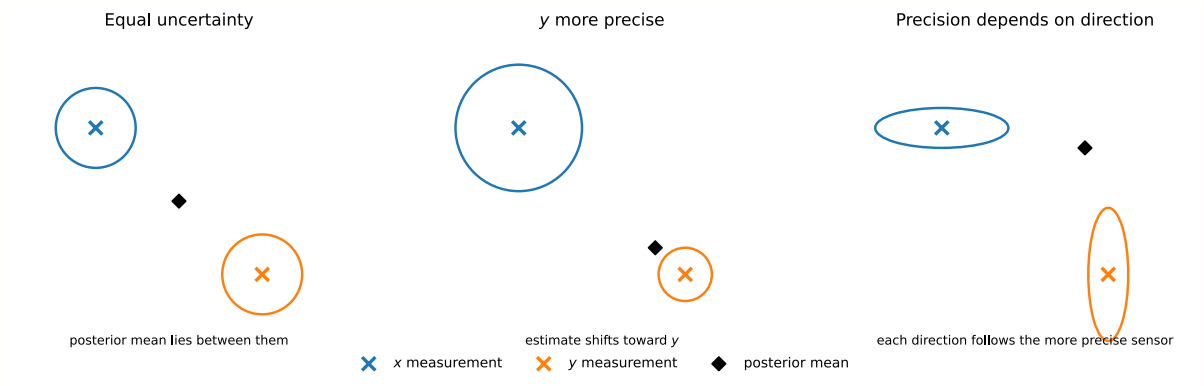
Sensor fusion combines measurements according to their uncertainty

Two noisy measurements of one state

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)\; x \mid z \sim \mathcal{N}(z, \Sigma_x), \quad y \mid z \sim \mathcal{N}(z, \Sigma_y)$$

Same two operations

$$p(z \mid x, y) \quad \rightarrow \quad E[z \mid x, y]$$



Qualitative weak-prior-limit schematic after Murphy, Book 2 §2.3.2.3, Fig. 2.10; `sensor_fusion_figure.ipynb`

- With independent sensor noises, equal isotropic noise gives the midpoint in the weak-prior limit; finite-prior mean and precision also matter.
- A more precise sensor pulls the posterior mean closer to its measurement
- With anisotropic uncertainty, the weighting can differ by direction

The same model can also predict measurements

Model and joint

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$
$$p(z, y) = p(z)p(y \mid z)$$

Marginalization rule

$$p(y) = \int p(z, y)\, \mathrm{d}z$$
$$= \int p(y \mid z)p(z)\, \mathrm{d}z$$

$$p(y) = \mathcal{N}\left(y \mid \mu_y, \Sigma_{yy}\right) \mu_y = W\mu_z + b, \quad \Sigma_{yy} = W\Sigma_z W^T + \Omega$$

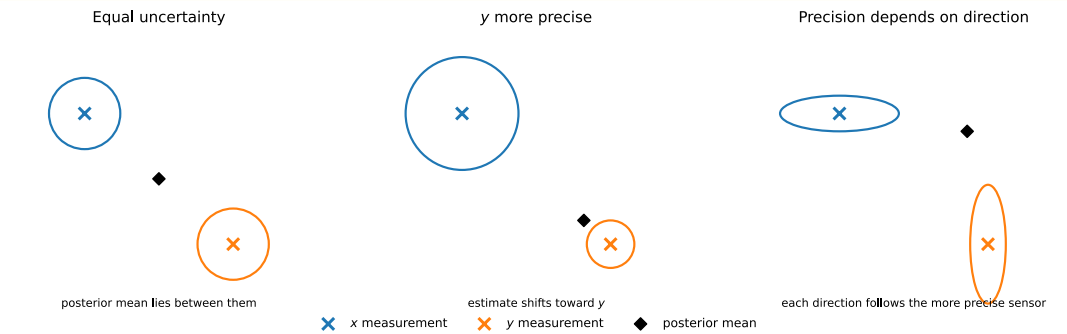
Murphy, Book 2 §2.3.2.2, Eq. 2.129

- Before observing y, marginalization gives the distribution of possible measurements
- These are the same μ_y and Σ_{yy} that appear inside the conditioning update

Summary: Linear Gaussian systems

Linear Gaussian model

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$
$$p(z, y) = p(z)p(y|z)$$



TAKE HOME

- The model plus affine-Gaussian rules determine the full joint distribution
- Posterior estimate = prior estimate + gain × prediction error
- More measurement noise gives smaller K; more prior uncertainty gives larger K

Lecture 1: four visual anchors

One dataset, several questions

A dataset is not a task.



$x \to \hat{y}$ Prediction
$x \to \hat{z}$ Discovery
$x_* \sim p(x \mid c)$ Generation

Three probability operations

The task determines the operation.

Summarize $E[g(X)] = \int g(x)p(x)\,\mathrm{d}x$
Marginalize $p(x) = \int p(x,y)\,\mathrm{d}y$
Condition $p(x \mid y) = \frac{p(x,y)}{p(y)}$

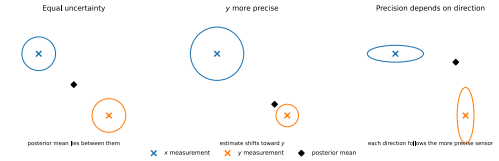
Gaussian conditioning

Observation changes estimate and uncertainty.

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Sensor fusion

Uncertainty controls how evidence is combined.



Visual anchors from Lecture 1: MNIST tasks, three probability operations, Gaussian conditioning, and sensor fusion